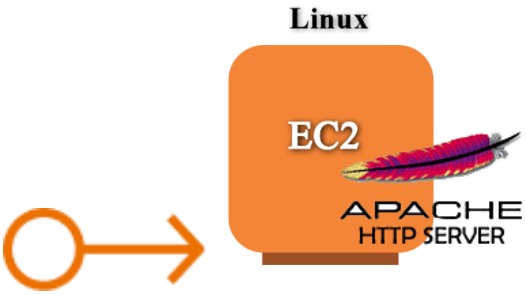
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

6.Configure Elastic IP Address to Windows Web Server in AWS



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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

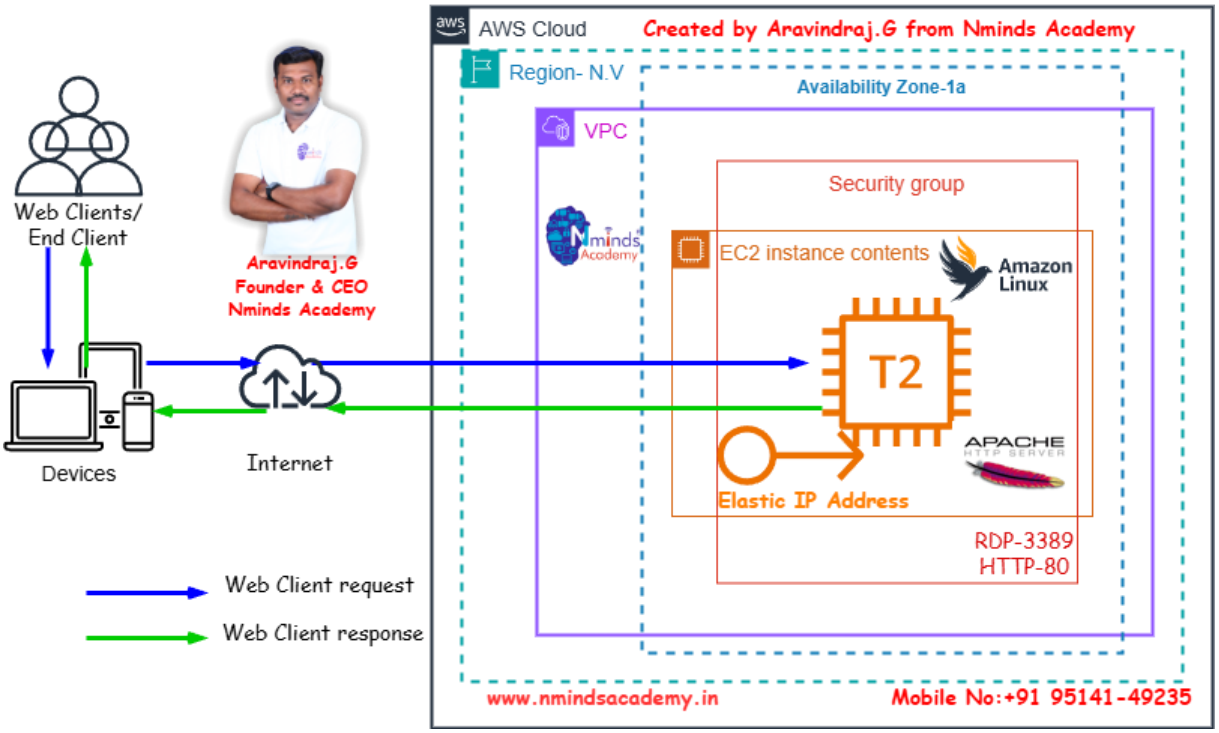
1. Highly Available Applications:
   - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
   - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
   - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



Used Services in AWS

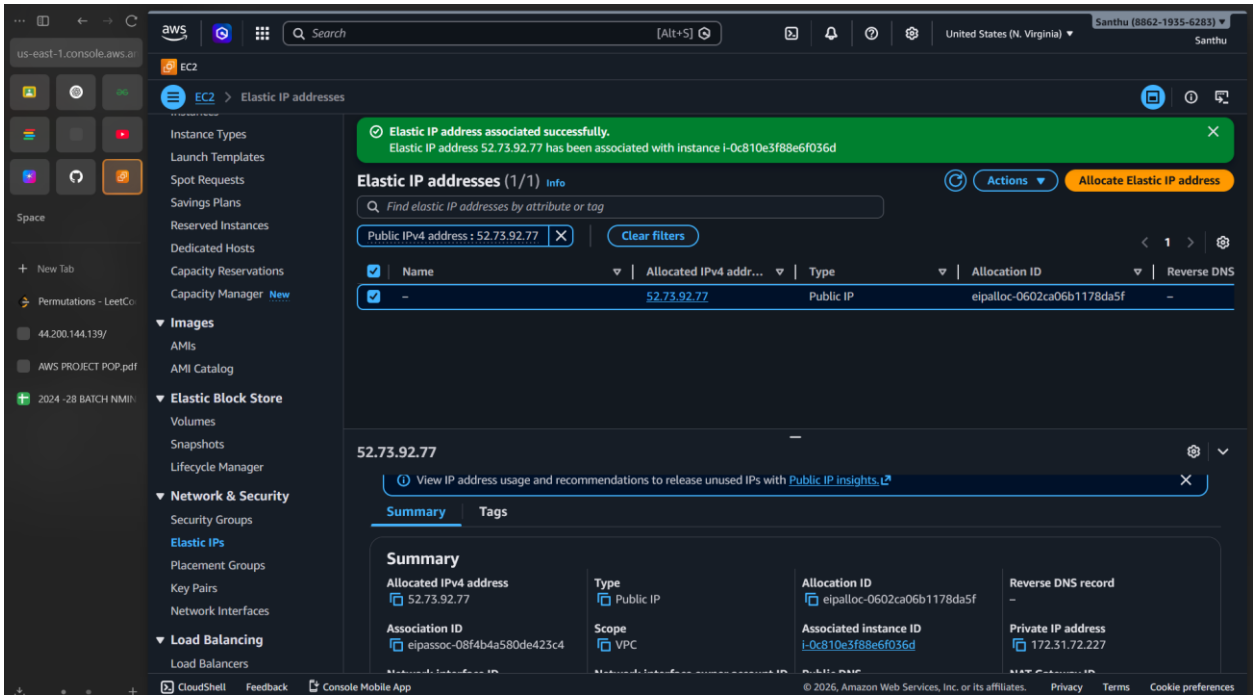
Used Services in Windows Server

Amazon EC2

APACHE HTTP SERVER

Apache Webserver

Amazon Elastic IP Address


Elastic IP address associated successfully.
Elastic IP address 52.73.92.77 has been associated with instance i-0c810e3f88e6f036d
Elastic IP addresses (1/1) Info
Actions
Allocate Elastic IP address
Find elastic IP addresses by attribute or tag
Public IPv4 address : 52.73.92.77
Clear filters
Name
Allocated IPv4 addr...
Type
Allocation ID
Reverse DNS
52.73.92.77
Public IP
eipalloc-0602ca06b1178da5f
52.73.92.77
View IP address usage and recommendations to release unused IPs with Public IP insights.
Summary
Tags
Allocated IPv4 address
52.73.92.77
Type
Public IP
Allocation ID
eipalloc-0602ca06b1178da5f
Reverse DNS record
Association ID
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Scope
VPC
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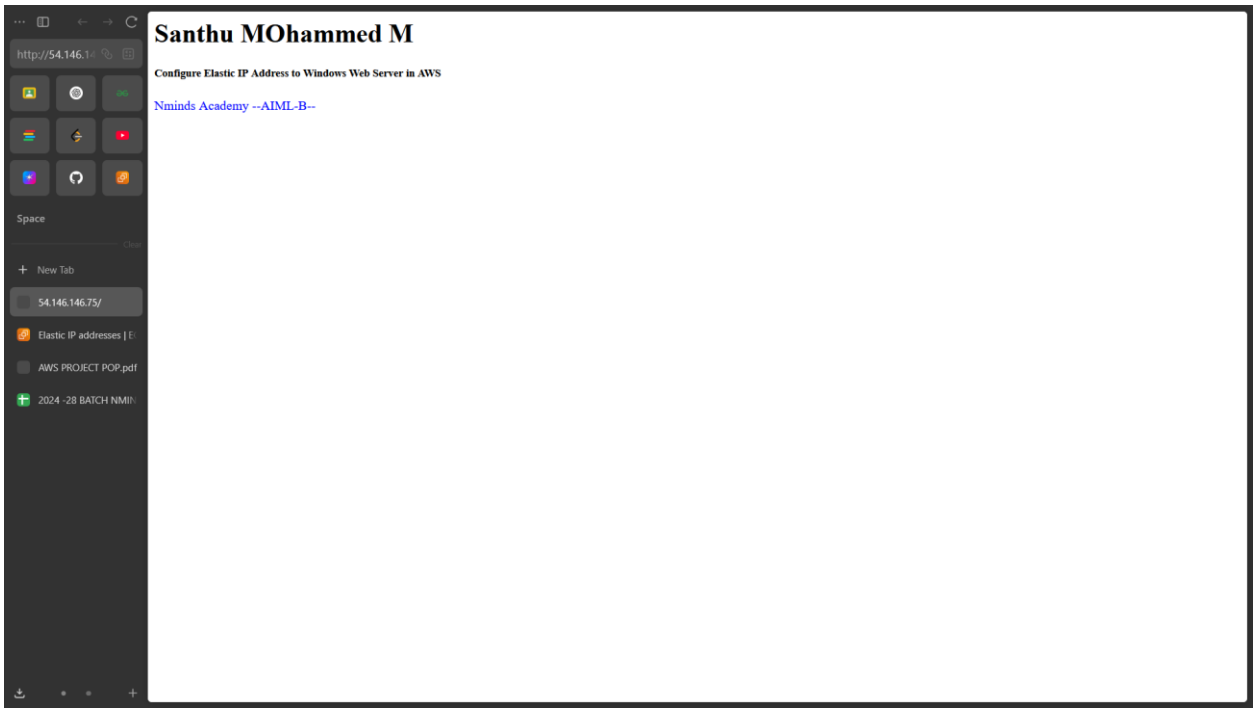
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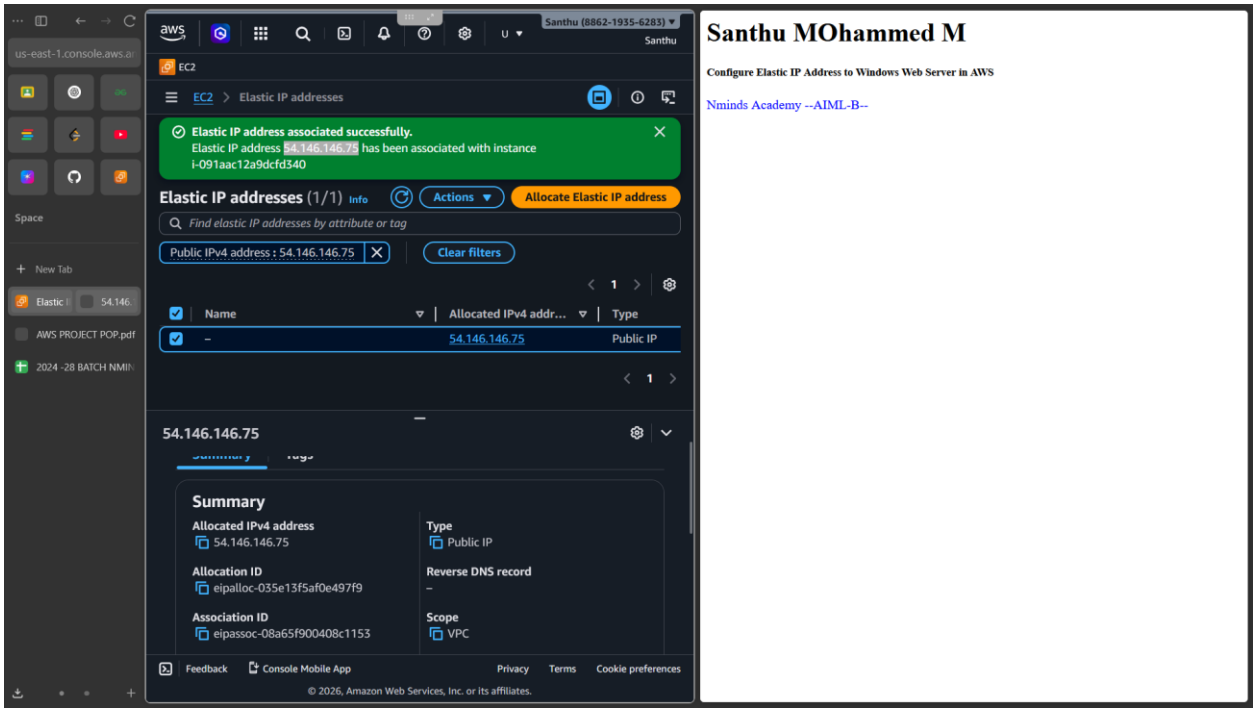
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Santhu
EC2
EC2 > Elastic IP addresses
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Elastic IP address 54.146.146.75 has been associated with instance i-091aac12a9dcfd340
Elastic IP addresses (1/1) Info
Actions
Allocate Elastic IP address
Find elastic IP addresses by attribute or tag
Public IPv4 address : 54.146.146.75
Clear filters
Name
Allocated IPv4 addr...
Type
54.146.146.75
Public IP
54.146.146.75
Summary
Allocated IPv4 address
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Scope
VPC
Feedback
Console Mobile App
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